

云原生平台下大语言模型PD分离架构的规模化挑战与实践

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- 03 RoleBasedGroup核心特性
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云原生平台部署大语言模型

背景

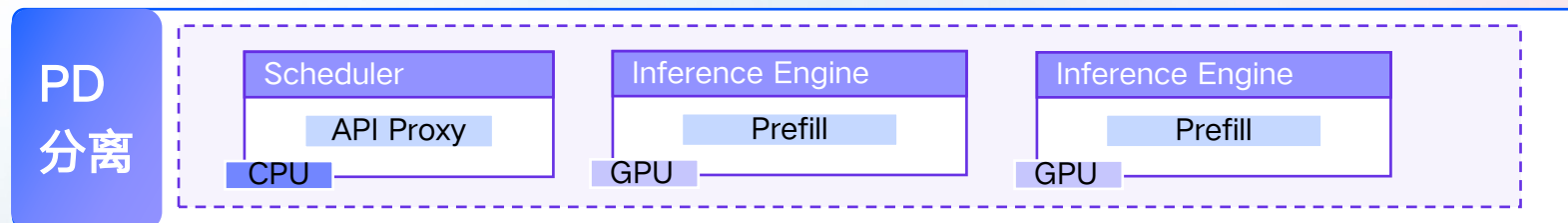
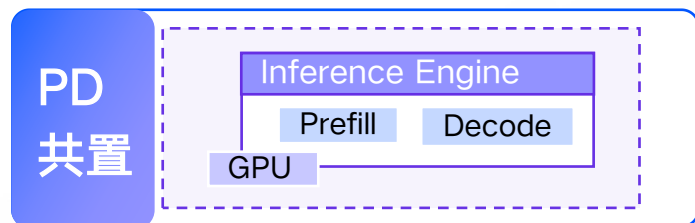
- **Prefill (预填充) – 理解问题阶段**

从用户输出的Prompt到生成第一个Token的过程

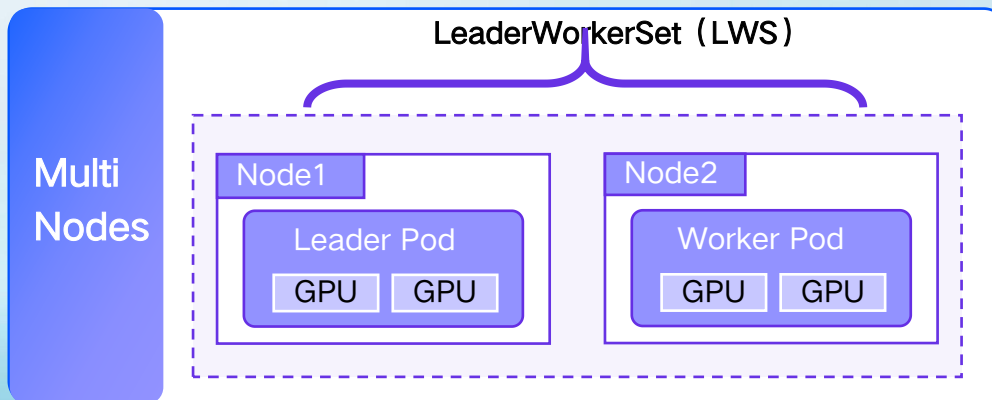
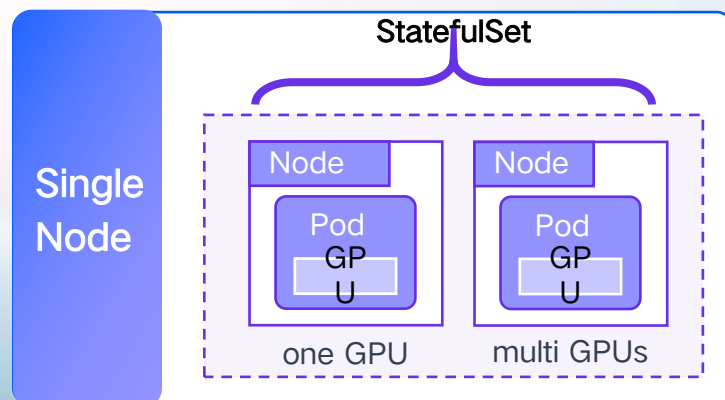
- **Decode (解码) – 逐字回答阶段**

生成第二个Token到最后一个Token的过程

- **部署架构**



- **部署常用工作负载**



云原生平台部署大语言模型

PD分离架构生产落地的问题与挑战

➤核心问题：PD分离架构的推理服务作为一个需要**多角色协同工作**的**有状态**应用，如何在处理Prefill和Decode在部署、运维过程中协作机制的同时，结合Router和KVCache Store提供一体化部署方案。

部署&运维复杂性

- 多角色间强顺序依赖
- 部署模板维护复杂
- 缺失角色间服务发现机制
- 缺失多角色协同伸缩机制
- 缺少一体化部署方案

资源动态性&有状态性

- 有状态应用迁移、故障重建、升级成本高
- 传统弹性指标失准
- 缺少KVCache Aware的拓扑调度机制

运行可靠性

- 多角色协同升级、回滚操作困难，需要人工介入
- 灰度过程流量验证困难
- 缺少多级故障恢复机制，存在及联故障雪崩风险

RoleBasedGroup

简介

➤ RBG是SGLang社区开源的一种面向多角色协同工作场景设计的工作负载，旨在解决**多角色场景**下的生命周期管理难题，包括多角色的创建、调度、升级、故障自愈、服务发现等能力。

Extensive

可扩展的引擎接入

可扩展的角色定义

Stable

原地升级

多级故障恢复

Coordination

协同升级

协同扩容

协同调度

Auto Discovery

托管服务发现

服务注册

Placement

Gang调度

拓扑亲和性调度

Elastic Scale

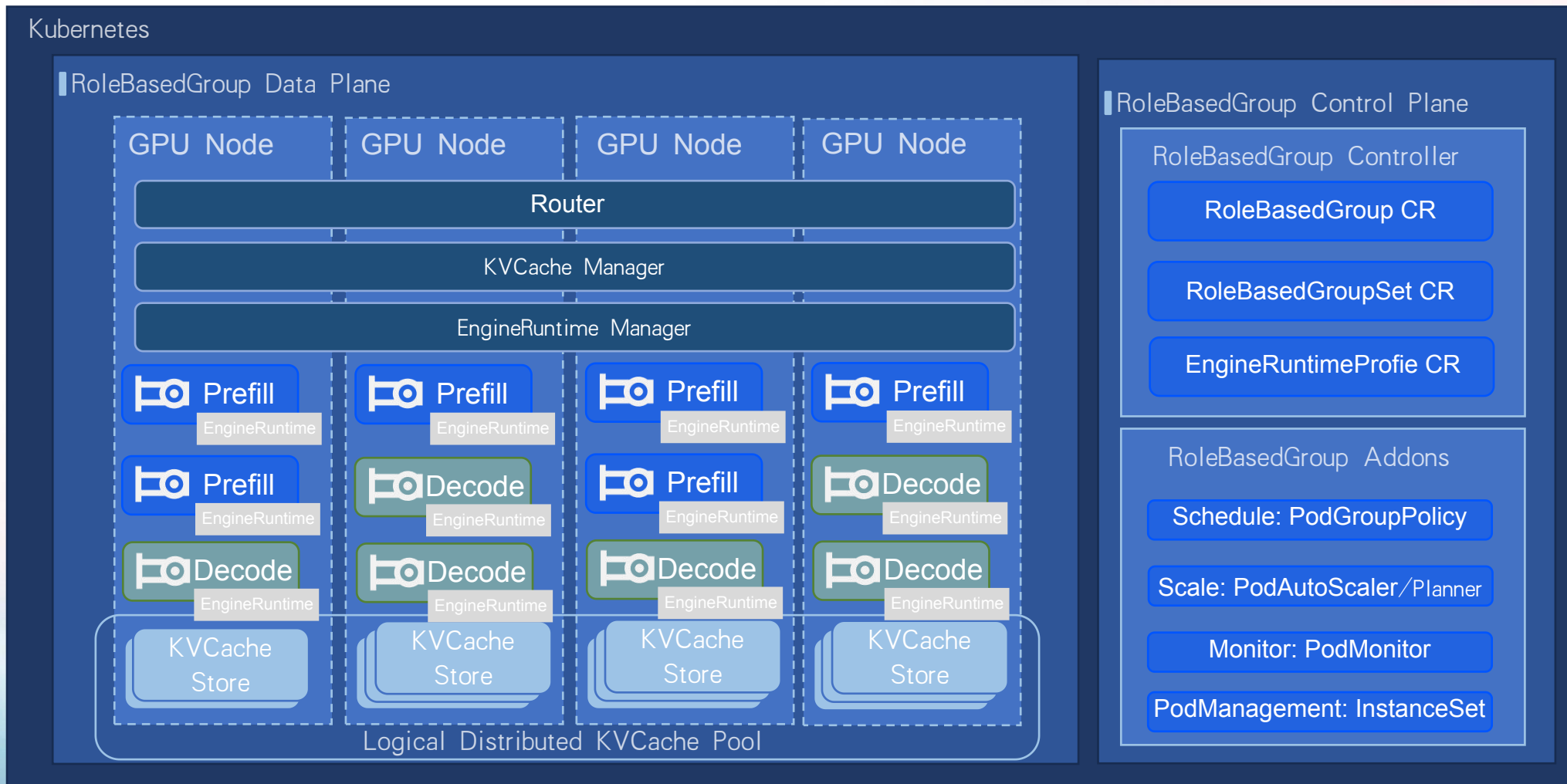
角色独立伸缩

资源智能规划

项目地址：<https://github.com/sgl-project/rbg>

RoleBasedGroup

推理套件一体化部署



RoleBasedGroup核心特性

Extensive: 可扩展多角色定义

模版抽象 减少冗余定义

```
apiVersion: workloads.x-k8s.io/v1alpha1
kind: RoleBasedGroup
metadata:
  name: sglang-pd-demo
spec:
  roleTemplate:
    - name: engine
      template:
        spec:
          containers:
            - name: router
              command:
                - ...
              env:
                - name: DEMO_KEY
                  value: "demo_value"
              volumeMounts:
                - name: model
                  mountPath: /model
              volumes:
                - name: model
                  ...
```

自定义角色依赖与负载类型

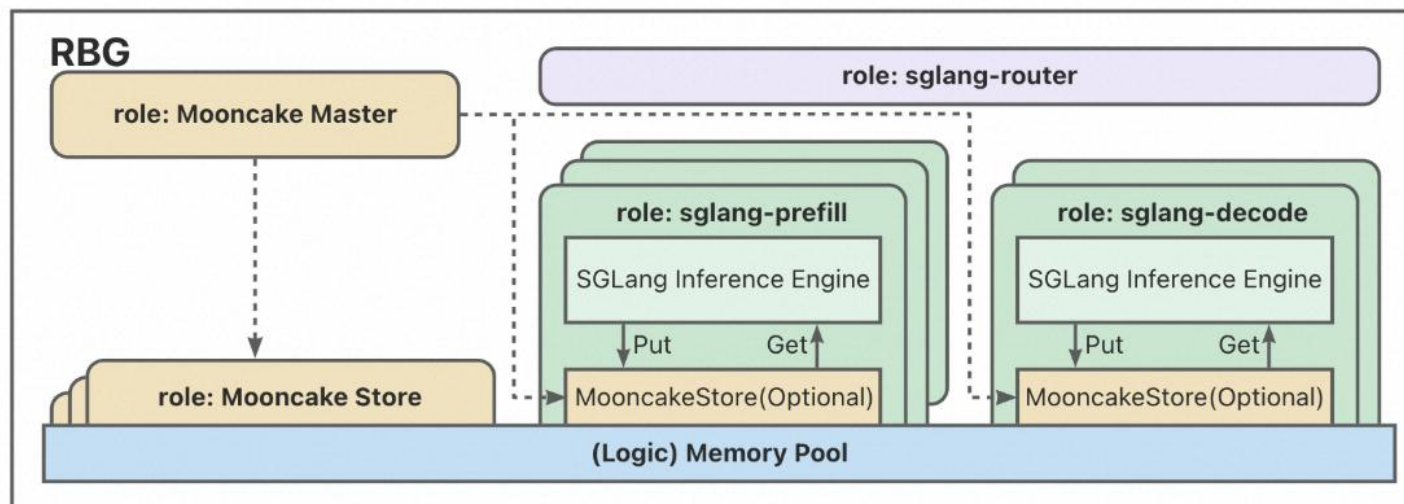
```
roles:
  - name: router
    replicas: 1
    dependencies: ["prefill", "decode"]
    workload:
      apiVersion: apps/v1
      kind: statefulset #Custom Defined
      template:
        spec:
          containers:
            - name: router
              command:
                - ...
              env:
                image: router:latest
```

EngineRuntime灵活注入

```
- name: prefill
  replicas: 2
  templateRef: engine
  templatePatch:
    spec:
      containers:
        - name: engine
          command:
            - ... --prefill
      engineRuntimes:
        - profileName: patio-runtime
- name: decode
  replicas: 2
  templateRef: engine
  templatePatch:
    spec:
      containers:
        - name: engine
          command:
            - ... --decode
      engineRuntimes:
        - profileName: patio-runtime
```

RoleBasedGroup核心特性

Extensive: Mooncake Integration



RoleBasedGroup核心特性

Stable: 滚动升级&原地升级

```
roles:
- name: prefill
  replicas: 3
  rolloutStartegy:
    rollingUpdate:
      maxUnavailable: 2
      maxSurge: 2
      updateStrategy: InplaceIfPossible
```

RBG

Role-A

Role-B

```
metadata:
  name: pod-t25zq
  uid: xxx
spec:
  containers:
  - name: foo
    image: foo:v1
  - name: bar
    image: bar:latest
  nodeName: node-A
status:
  podIP: 192.168.198.50
```

```
metadata:
  name: pod-t25zq
  uid: xxx
spec:
  containers:
  - name: foo
    image: foo:v2
  - name: bar
    image: bar:latest
  nodeName: node-A
status:
  podIP: 192.168.198.50
```

原地升级保证对于镜像版本的更新仅重启容器
升级后网络、拓扑等信息不变

	R-0	R-1	R-2	R-3	R-4	R-5
Stage1						
Stage2						
Stage3						
Stage4						
Stage5						
Stage6						
Stage7						
Stage8						
Stage9						

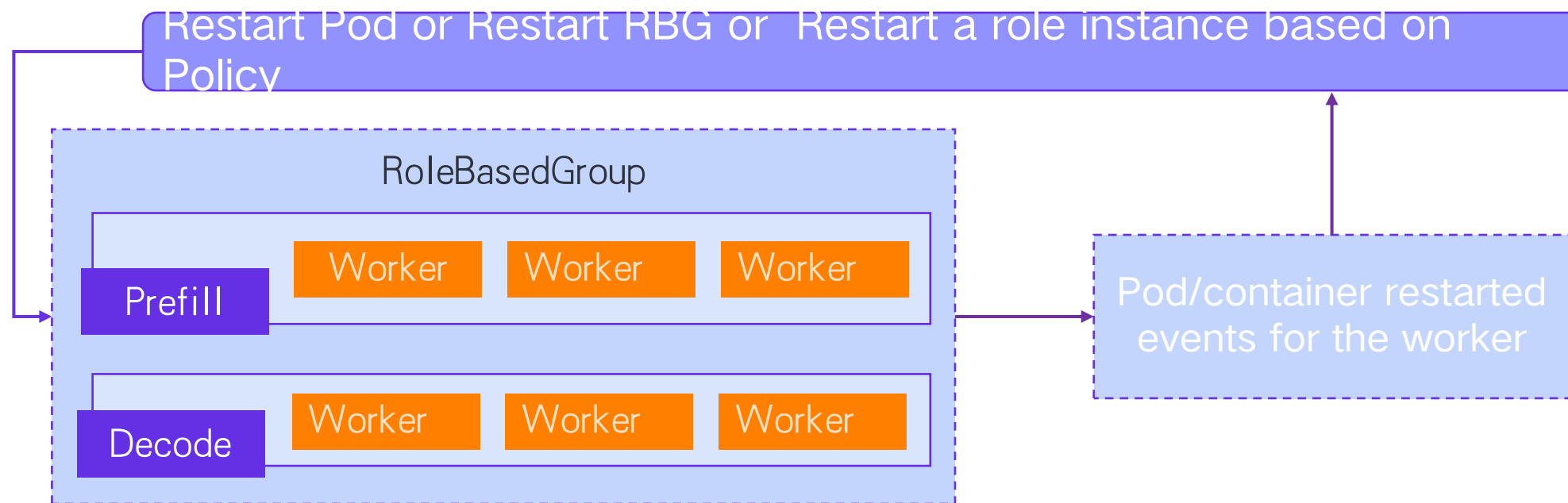
Replica has been updated
Replicas in rolling update

Replica hasn't been updated

RoleBasedGroup核心特性

Stable: 故障自愈&原地恢复

Configurable **restartPolicy**: Recreate/**InPlace Restart** Entire Role, Restart Entire Group, or Manual Control.



RoleBasedGroup核心特性

Coordination: 多角色协作更新

coordination:

- name: prefill-decode-update

type: RollingUpdate

roles:

- prefill

- decode

strategy:

maxUnavailable: 20%

maxSkew: 5%

partition: 20%

Stage 1:

Prefill Partition: 75% > partition

Decode Partition: 75% > partition

Current Skew:

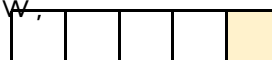
$Abs(4/20 - 1/5) = 0\% < maxSkew$,

更新继续

Prefill: 4 replicas in Updating / 20 replicas total



Decode: 1 replicas in Updating / 5 replicas total



Stage 2:

Prefill Partition: 50% > partition

Decode Partition: 60% > partition

Current Skew:

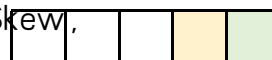
$Abs(10/20 - 2/5) = 10\% > maxSkew$,

此轮仅更新2个Prefill副本

Prefill: 6 replicas updated / 4 replicas in Updating / 20 replicas total



Decode: 1 replicas updated / 1 replicas in Updating / 5 replicas total



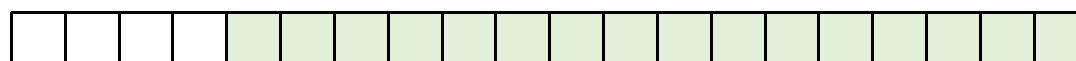
Stage 3:

Prefill Partition: 20% = partition

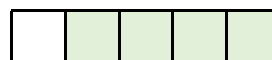
Decode Partition: 20% = partition

更新暂停

Prefill: 16 replicas updated / 20 replicas total

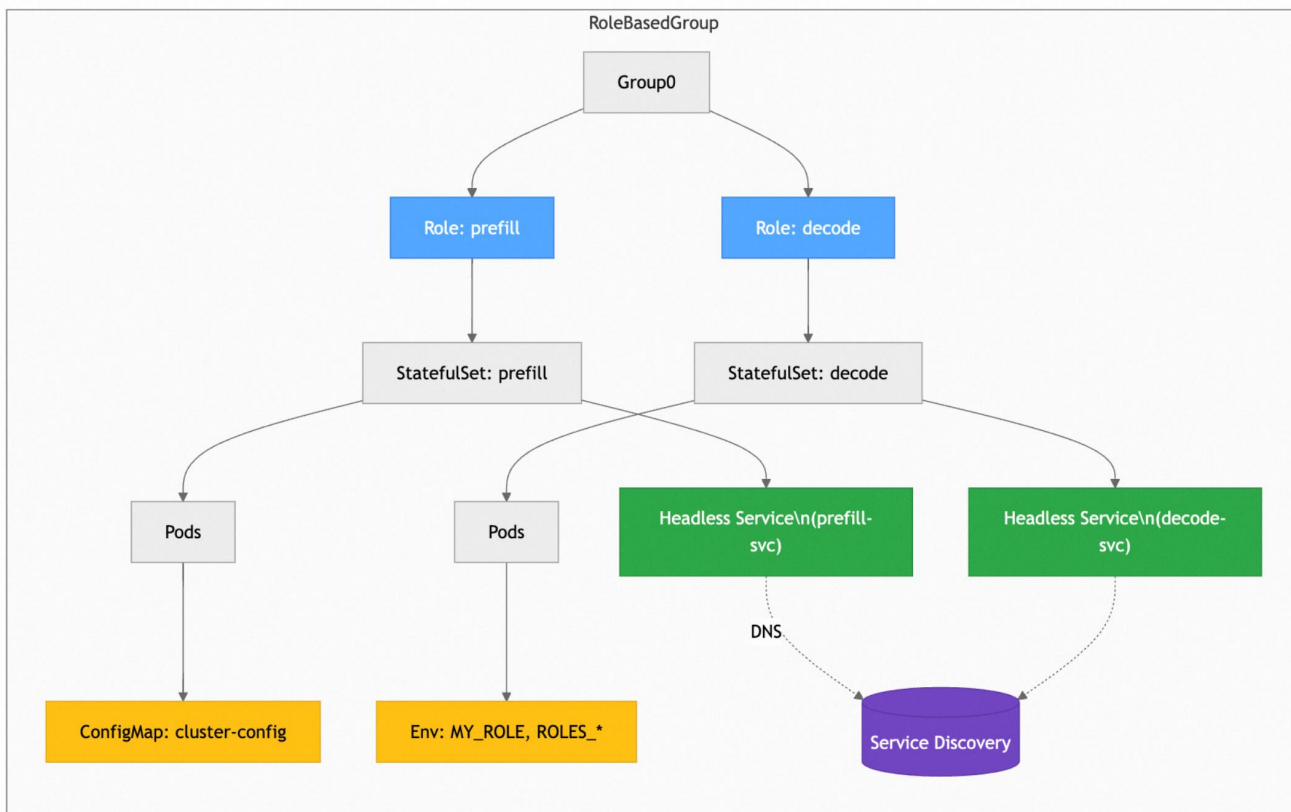


Decode: 4 replicas / 5 replicas total



RoleBasedGroup核心特性

Auto Service Discovery: 自动服务发现



自动注入全局拓扑信息
配置文件

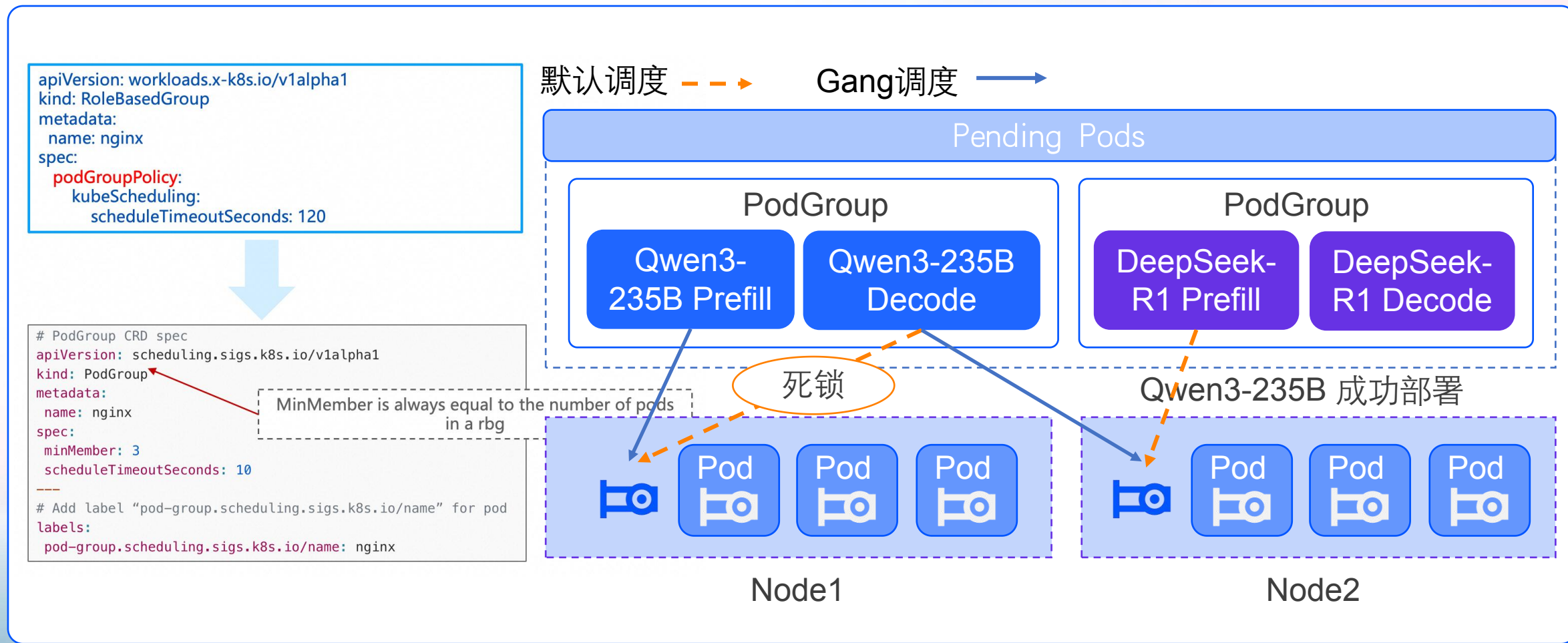
```
# /etc/rbg/config.yaml
group:
  name: sglang-pd-demo
  roles:
    - prefill
    - decode
  size: 2
roles:
  prefill:
    instances:
      - address: prefill-0.sglang-pd-demo-prefill
        size: 1
  deccode:
    instances:
      - address: decode-0.sglang-pd-demo-decode
        size: 1
```

自动注入自身拓扑信息
环境变量

```
env:
  - name: GROUP_NAME
    value: "sglang-pd-demo"
  - name: ROLE_INDEX
    value: "0"
  - name: ROLE_NAME
    value: "prefill"
```

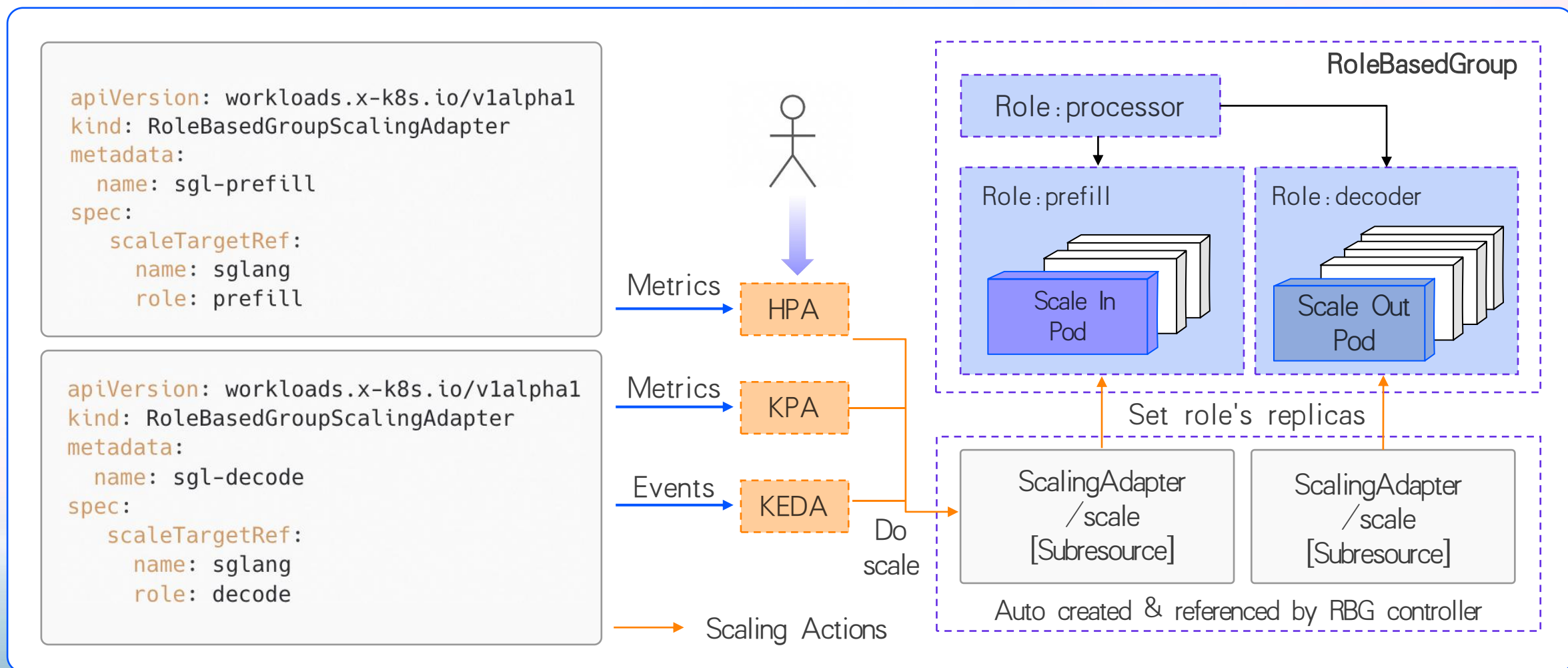
RoleBasedGroup核心特性

Placement: Gang调度



RoleBasedGroup核心特性

Elastic Scale: 角色独立伸缩配置

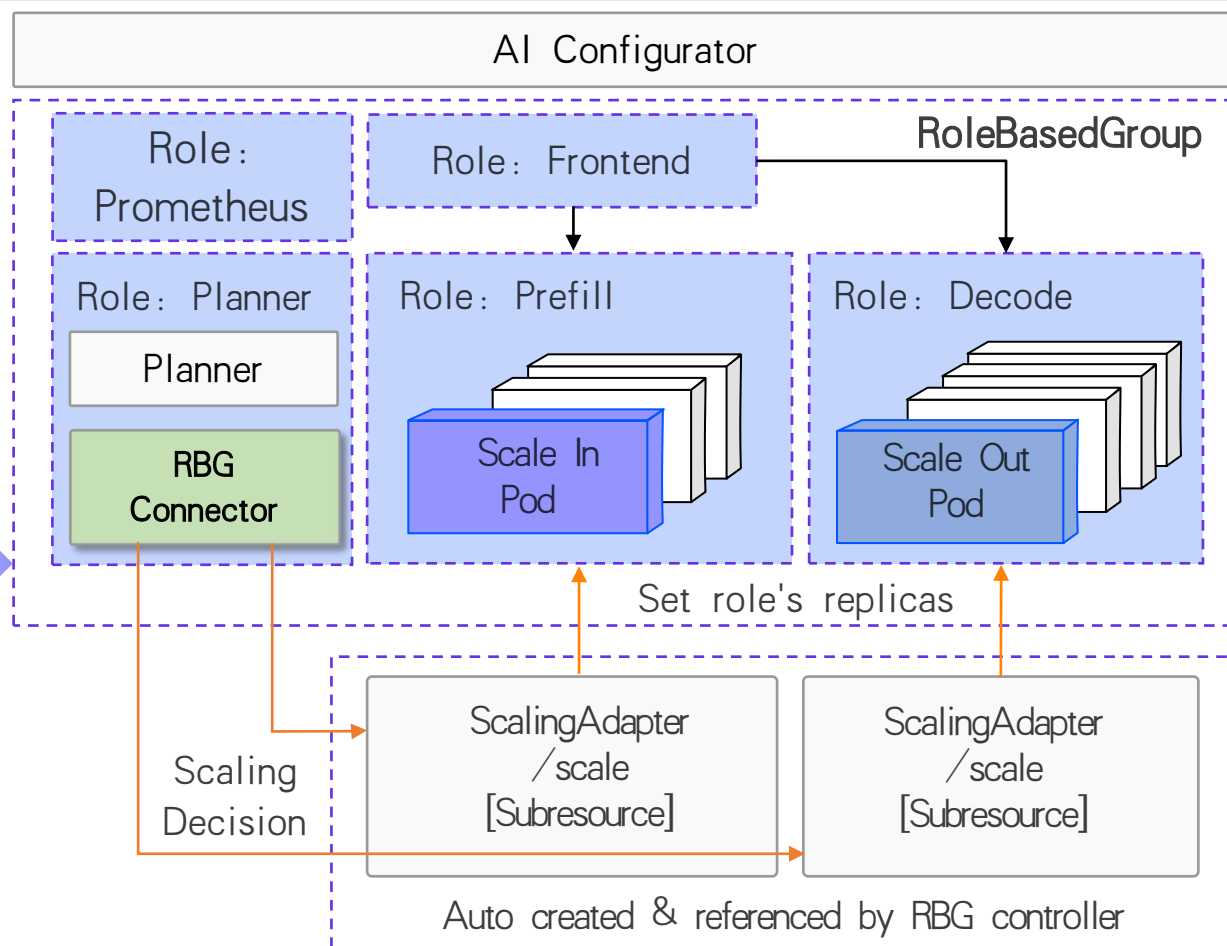
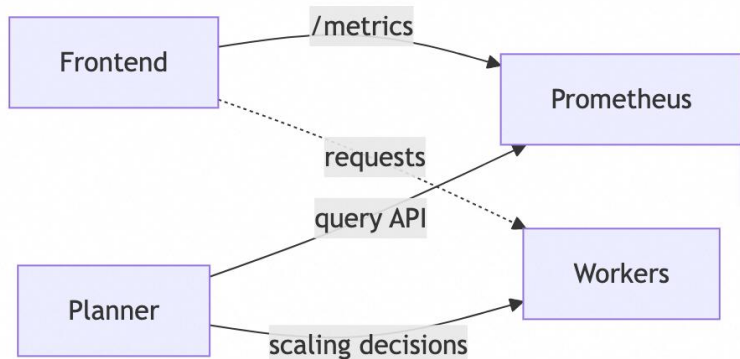


RoleBasedGroup核心特性

Elastic Scale: 资源智能规划

RBG & Dynamo社区最新合作

通过Dynamo Planner 实现SLA驱动的资源智能规划



Engine Runtime

为什么需要Engine Runtime

原因一：vLLM / SGLang等推理引擎的原地“状态”变化

- 动态模型参数更新能力 (例: `"/update_weights_from_tensor"`)
- 动态加载/卸载LoRA模型 (例: `"/load_lora_adapter"`)
- KV Cache的管理动作 (例: `"/flush_cache"`)

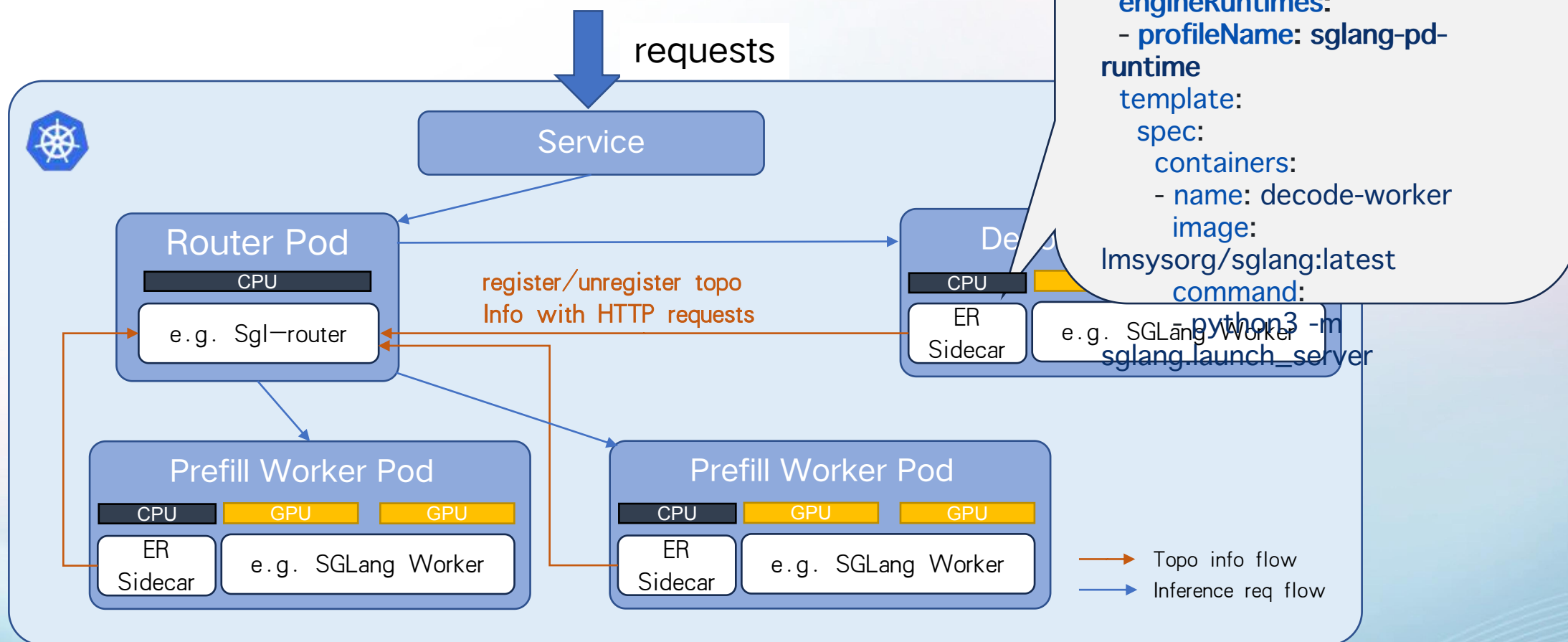
原因二：PD分离分布式推理的拓扑复杂性

- Prefill、Decode、Router多角色Pod间的拓扑发现
- 推理引擎拓扑的细粒度控制

```
apiVersion: workloads.x-  
k8s.io/v1alpha1  
kind: ClusterEngineRuntimeProfile  
metadata:  
  name: sglang-pd-runtime  
spec:  
  containers:  
    - image:  
      <ENGINE_RUNTIME_IMG>  
      name: engine-runtime  
    updateStrategy: NoUpdate
```

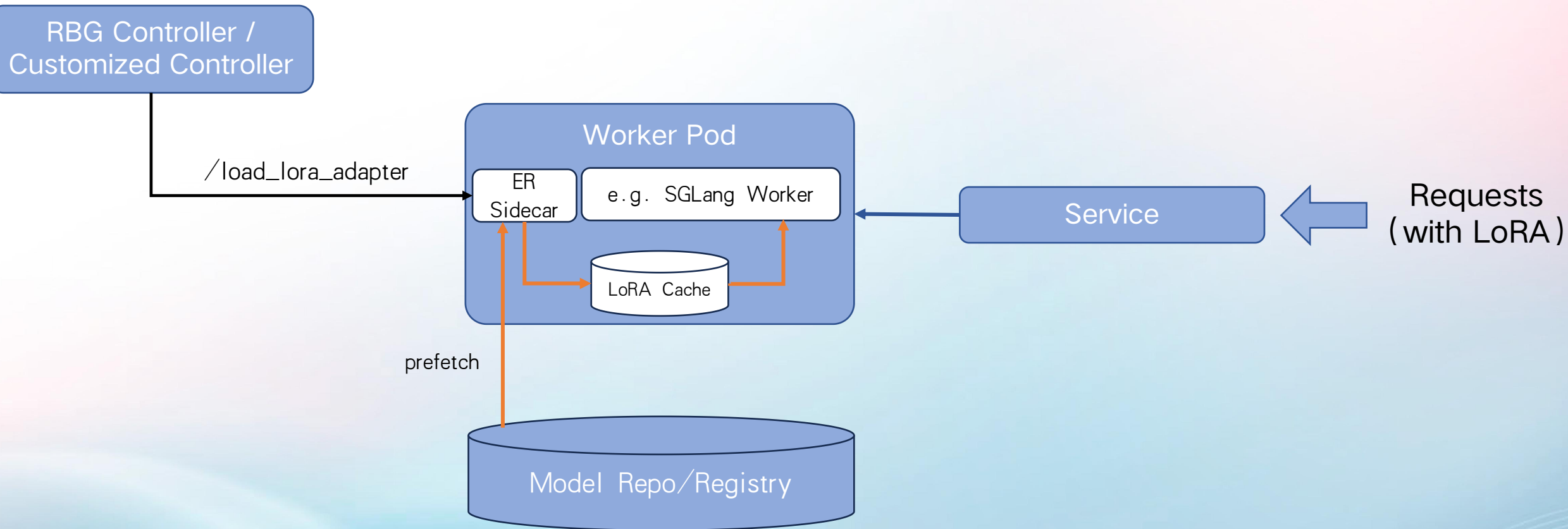
Engine Runtime

PD分离拓扑感知



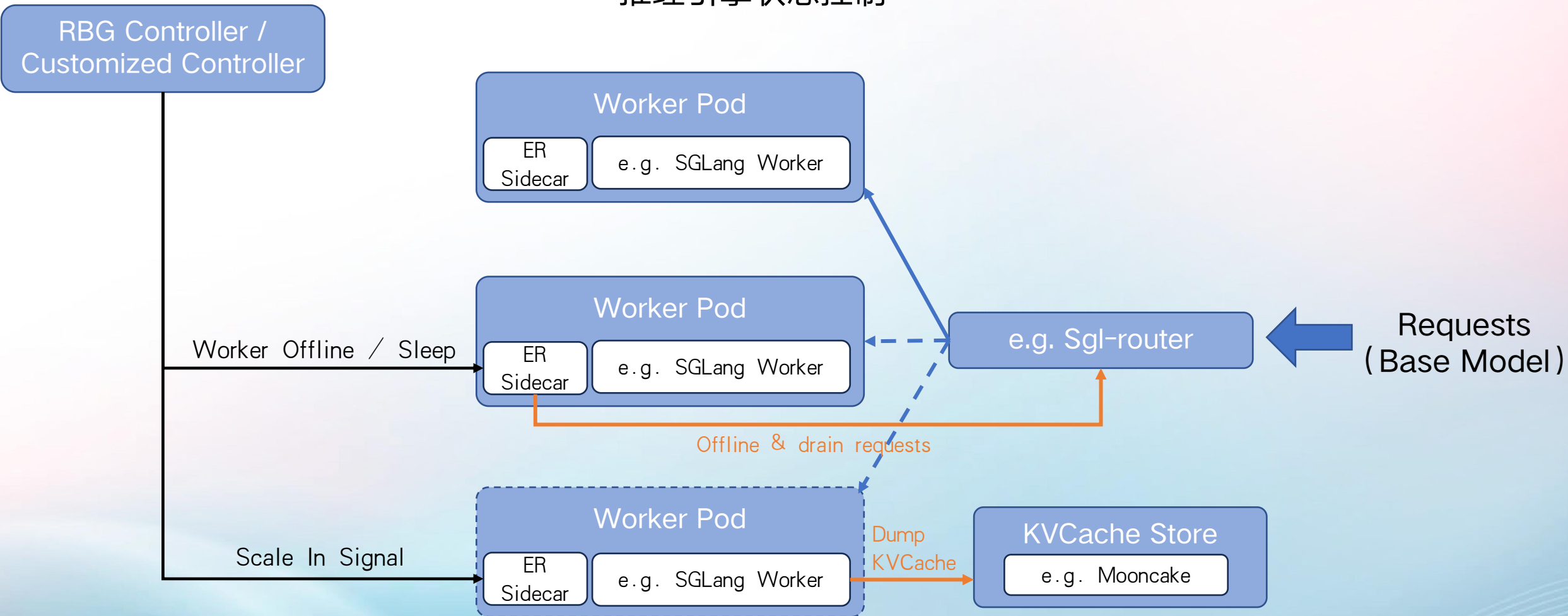
Engine Runtime

推理引擎LoRA模型管理



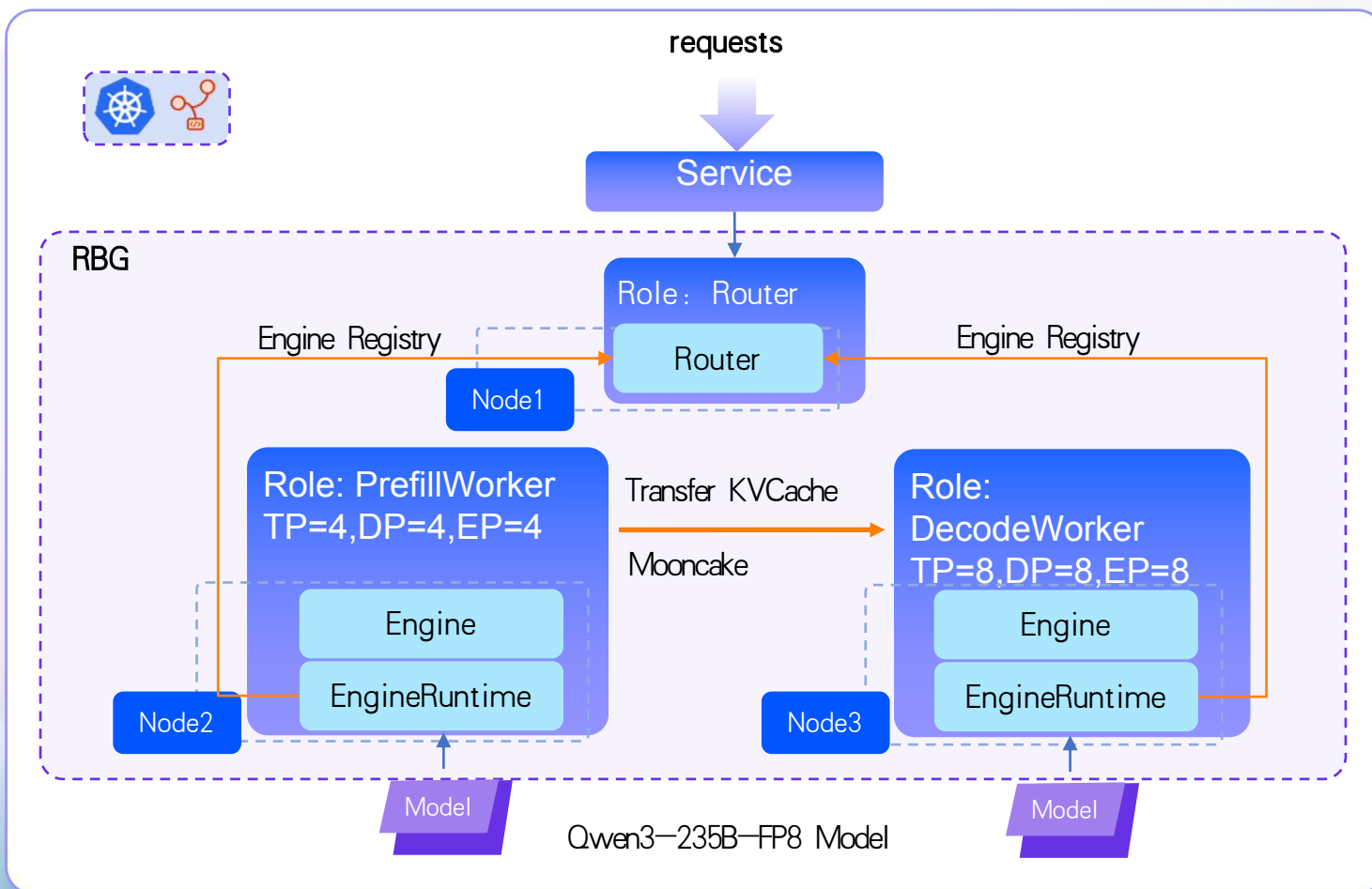
Engine Runtime

推理引擎状态控制

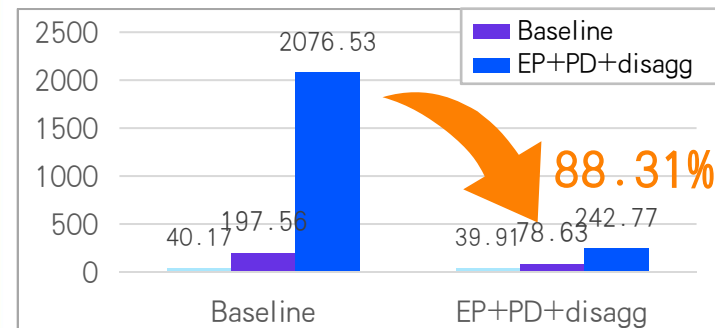


最佳实践

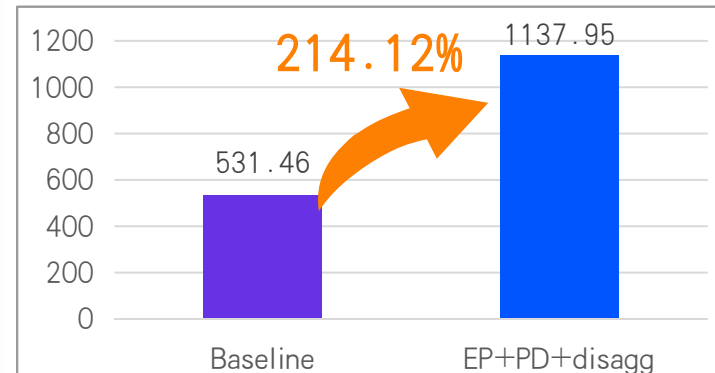
基于RBG部署PD分离架构Qwen3 235B模型



Baseline vs EP+PD+disagg ITL(ms)

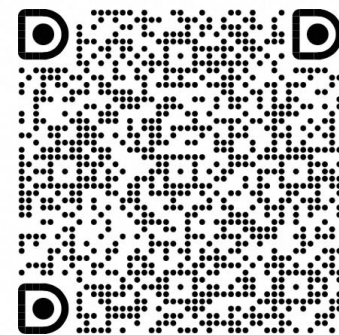
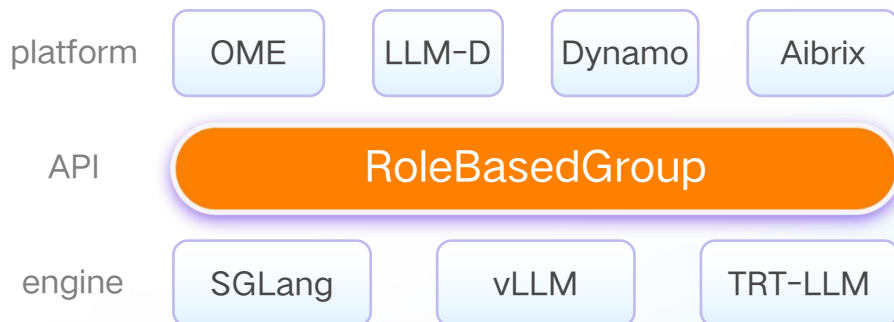


Baseline vs EP+PD+disagg Throughput



RBG 开源贡献 SGLang 社区

钉钉扫码加入SGLang RBG开源社区交流群



<https://github.com/sgl-project/rbg>

共建 K8s 管理大模型推理负载的标准 API



Dynamo



南京大学
NANJING UNIVERSITY